

Frequency Response of Amplifiers

FREQUENCY RESPONSE OF AMPLIFIERS

GBW Poles, Zeros

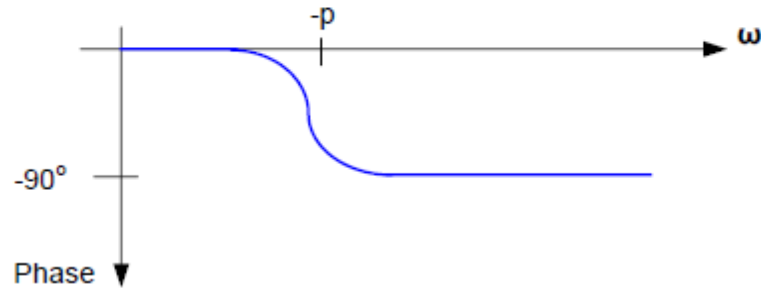
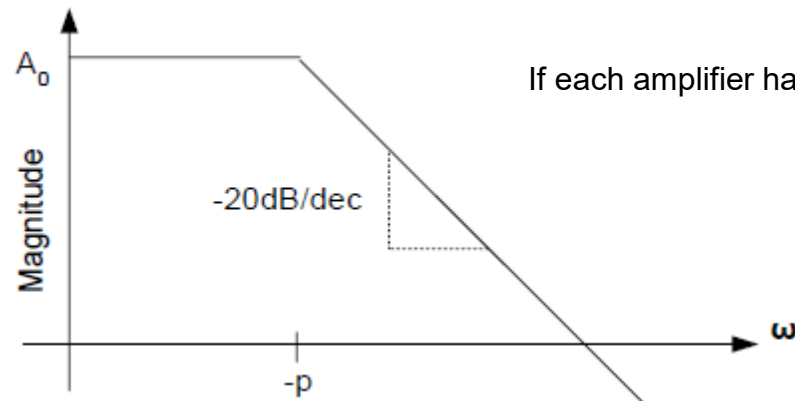
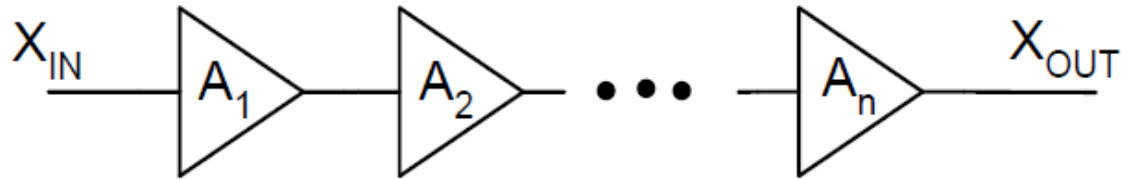
All digital and analog circuits
suffer from these parasitic
effects?

Switching loss limitation at
higher frequencies!!!!

Higher Current at higher
frequencies!!!

FREQUENCY RESPONSE OF AMPLIFIERS – GAIN-BANDWIDTH-PRODUCT

Cascading of Amplifiers

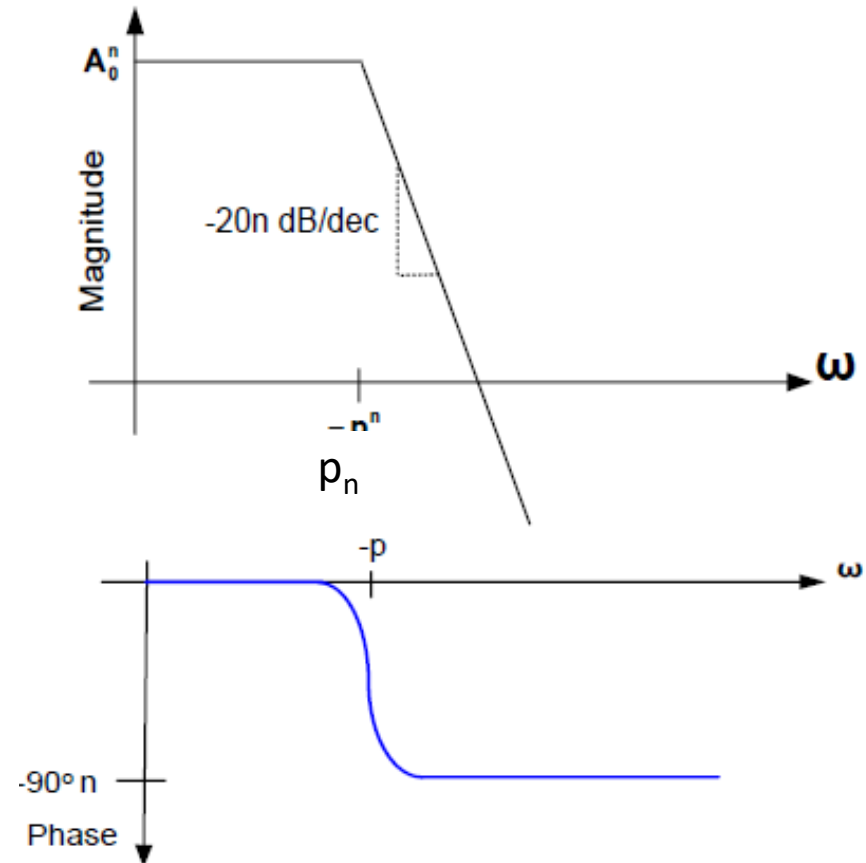


If each amplifier has low-pass characteristics...

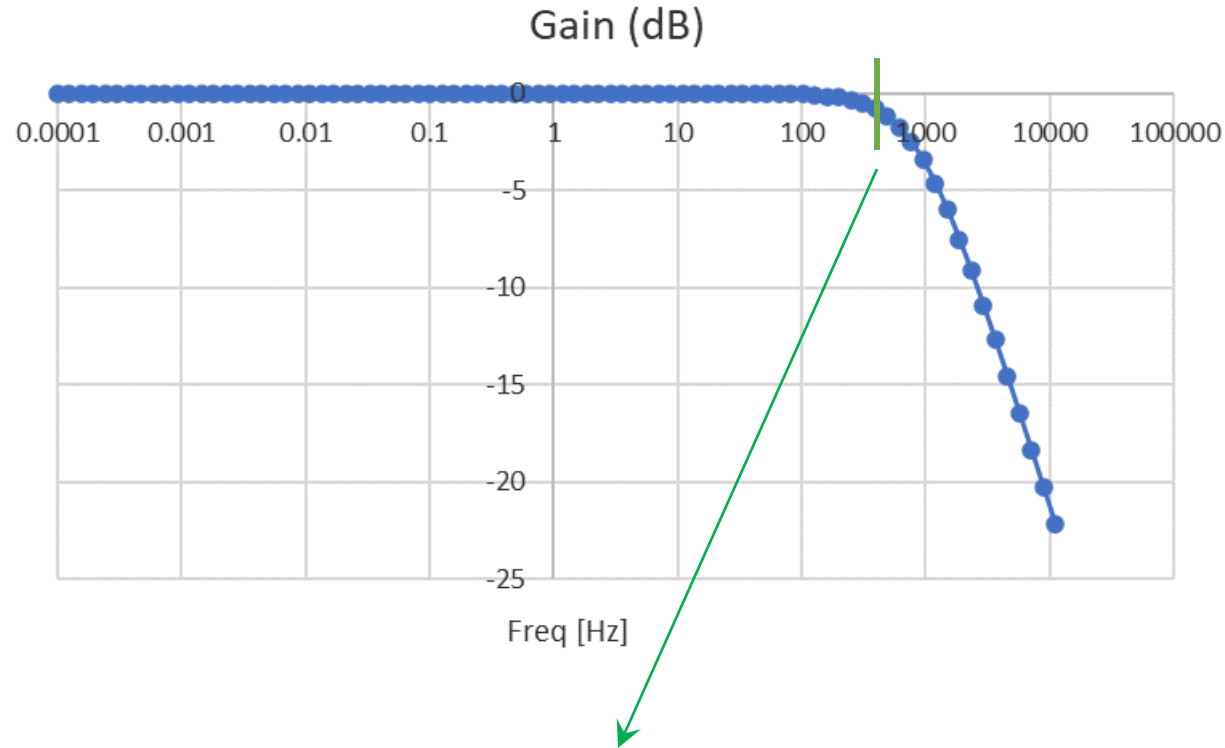


$$\frac{X_{OUT}}{X_{IN}} = A = \frac{\prod_{l=1}^n A_{0l}}{\prod_{k=1}^n \left(\frac{s}{\tilde{p}_k} + 1 \right)}$$

DC Gain



FREQUENCY RESPONSE OF AMPLIFIERS – GAIN-BANDWIDTH-PRODUCT



$$f_{3\text{-dB}} \approx 800 \text{ Hz}$$

$$\omega_{3\text{-dB}} \approx 5022 \text{ rad/s}$$

p_n is smaller than p . As a summary, cascading amplifier stages increases the total DC gain but reduces the total bandwidth of the cascaded system. However, for all scenarios, the gain-bandwidth-product remains constant.

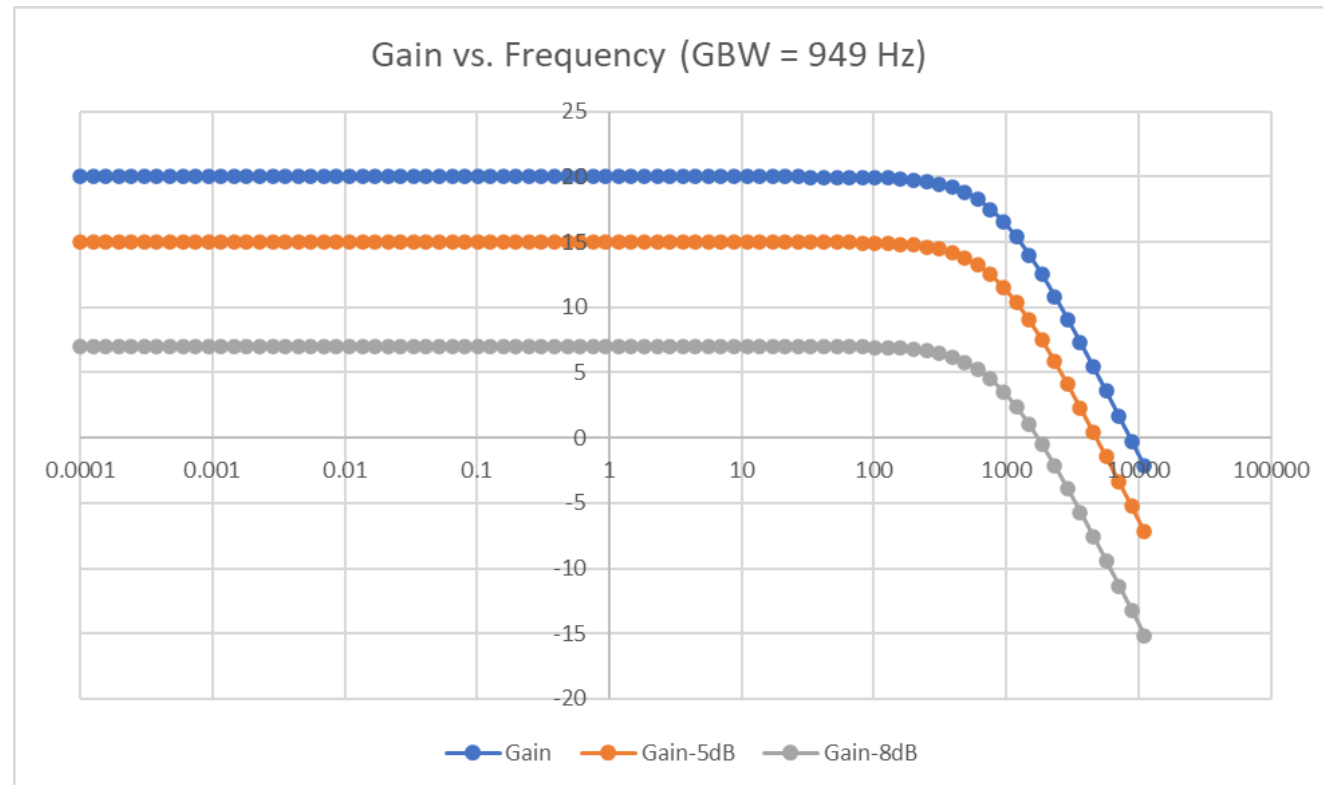
p is the pole of the amplifier. For a first-order amplifier or a filter, p is at the same time the angular frequency at which the voltage gain drops by 3 dB!

FREQUENCY RESPONSE OF AMPLIFIERS – GAIN-BANDWIDTH-PRODUCT

The Gain-Bandwidth-Product (GBW) is simply the product of scalar (not in dB) DC gain (A) and the 3-dB frequency (f_{3dB}) or in other words, the bandwidth.

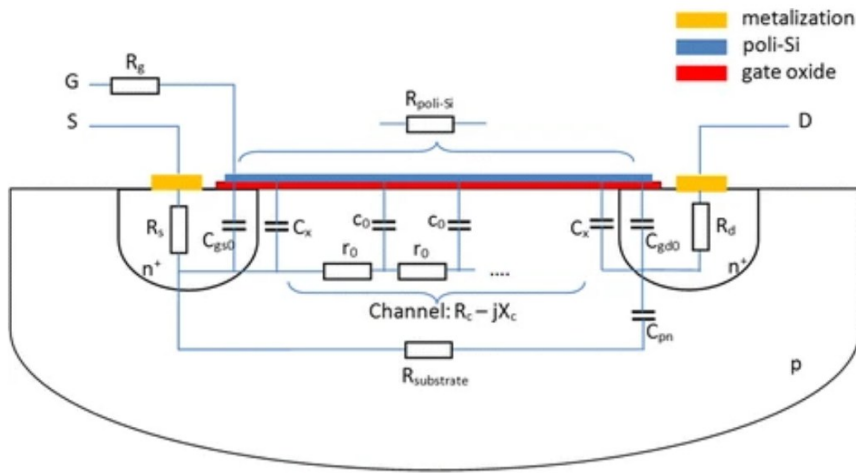
$$GBW = f_{3dB} \cdot A$$

If a higher bandwidth is needed, A needs to be reduced, for example by means of negative feedback. This is due to the fact that GBW is constant for a given circuit.

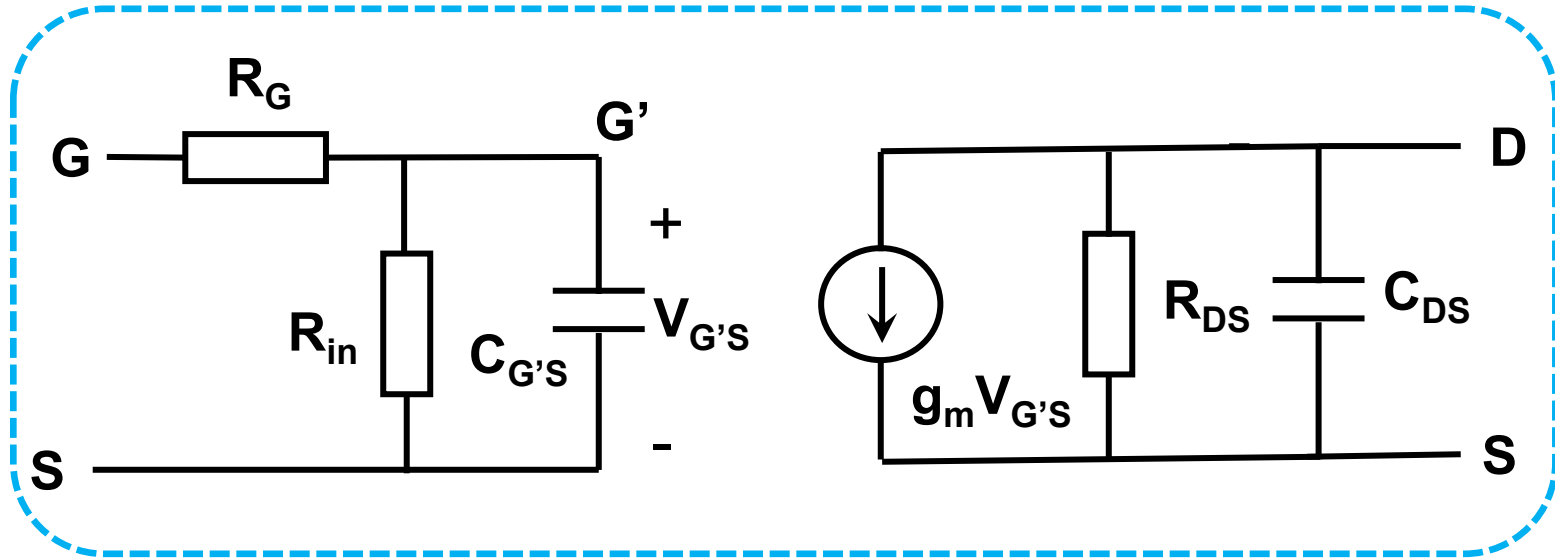


FREQUENCY RESPONSE OF AMPLIFIERS

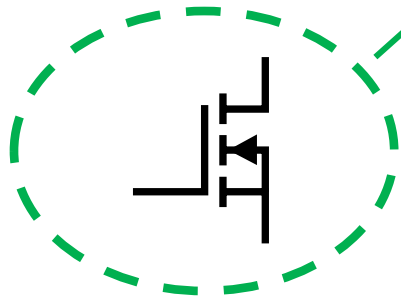
A MOSFET is not an ideal component and has parasitic capacitances and resistances. The MOSFET with the parasitic can be characterized as follows:



Simplified cross-section of MOSFET

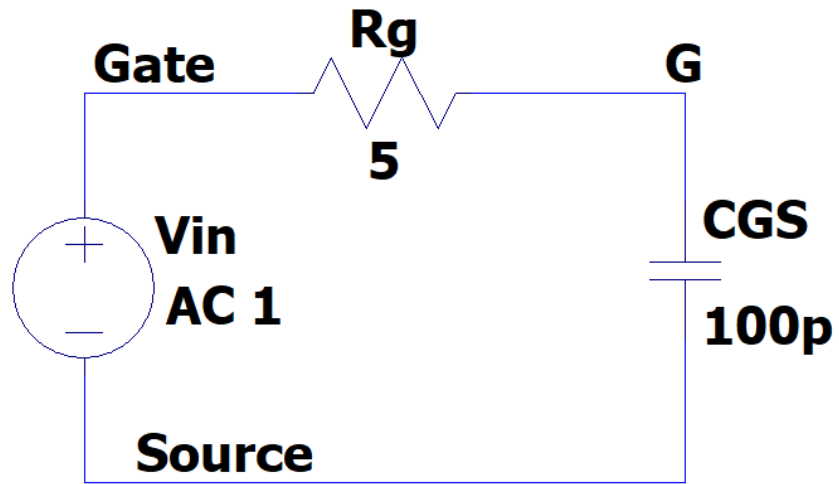


MOSFET small-signal equivalent circuit

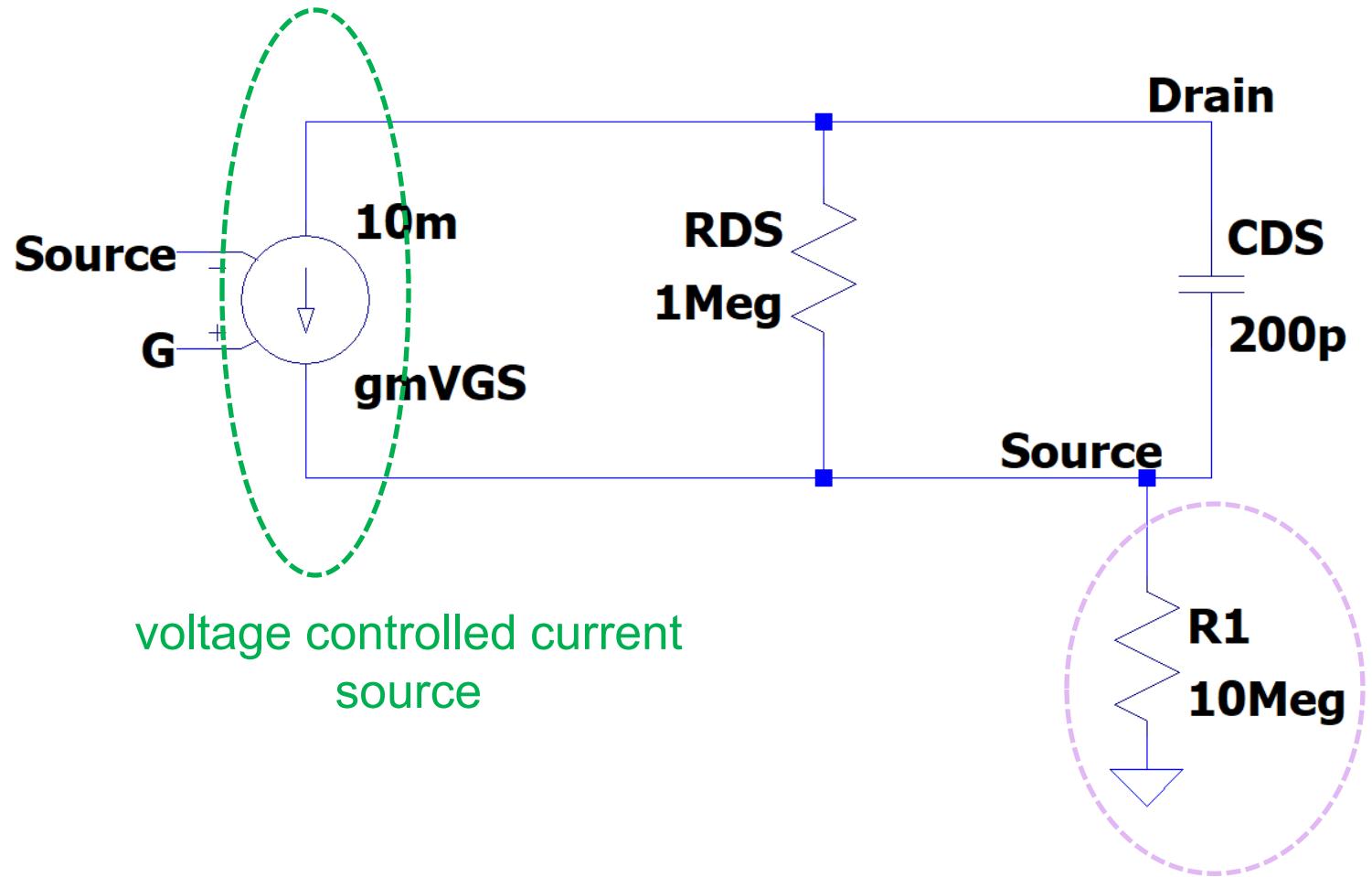


Simulation in LTSpice →

FREQUENCY RESPONSE OF AMPLIFIERS



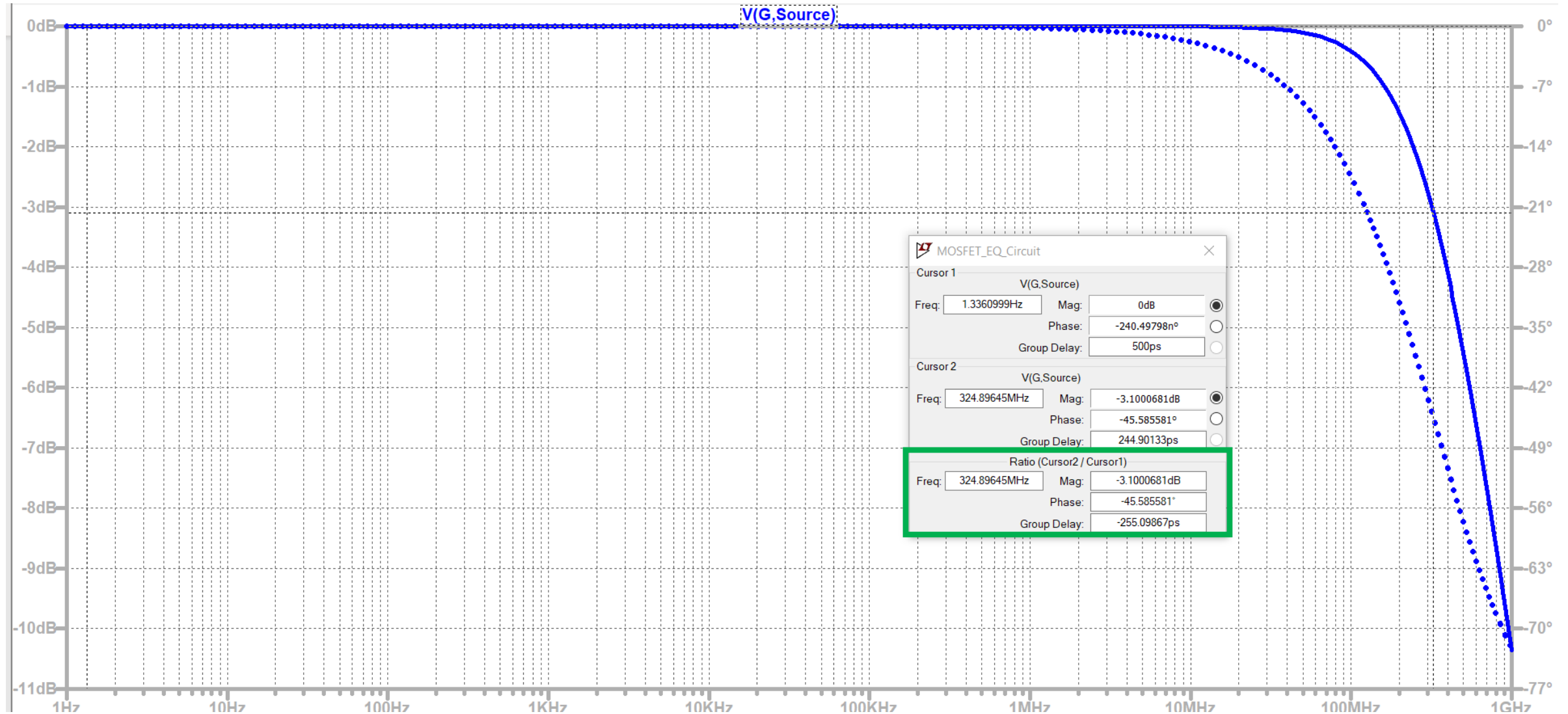
.ac dec 100 1 1G



* g_m and R_{DS} may not be realistic. These values have been selected to illustrate the frequency behavior of a transistor!

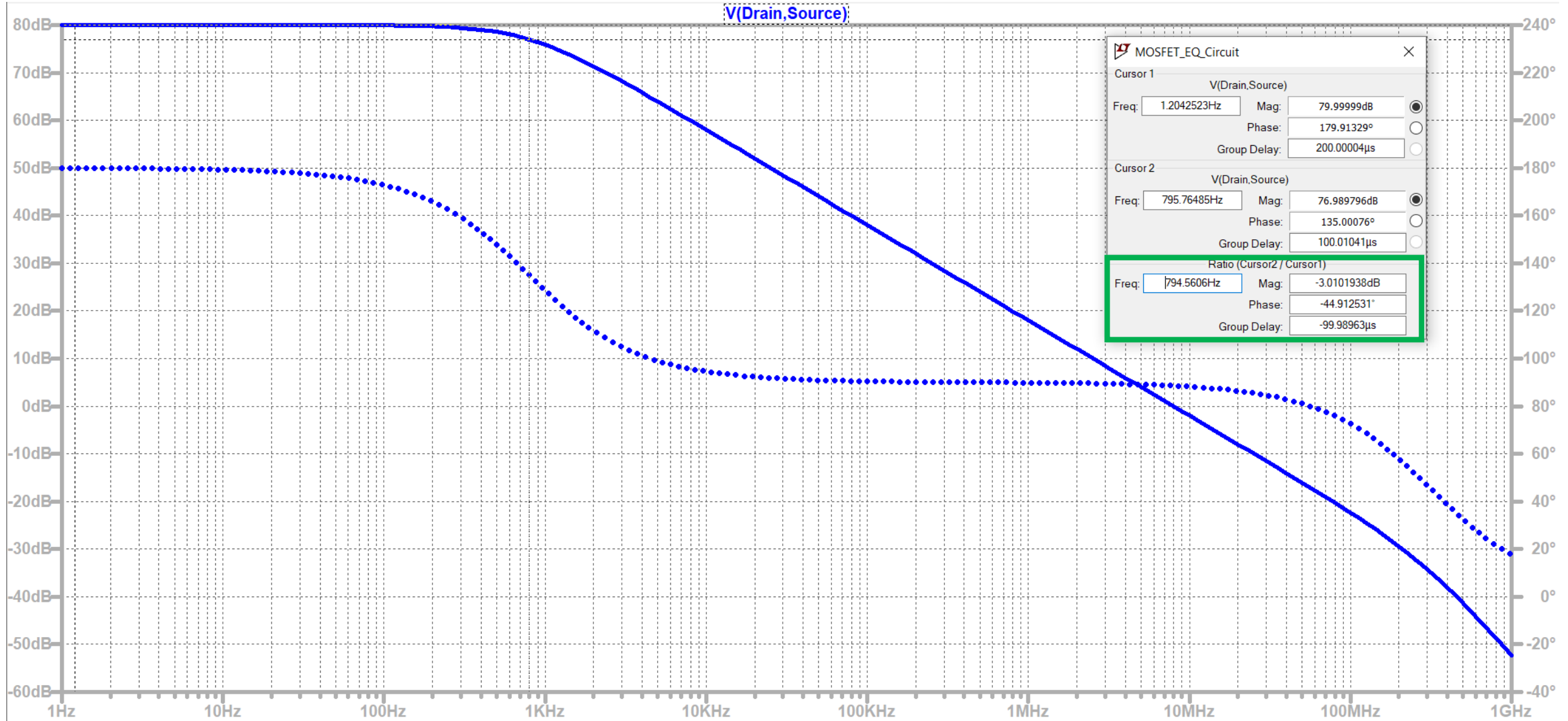
The resistor is necessary to have a ground connection in the simulation and does not have any impact on the circuit performance!

FREQUENCY RESPONSE OF AMPLIFIERS



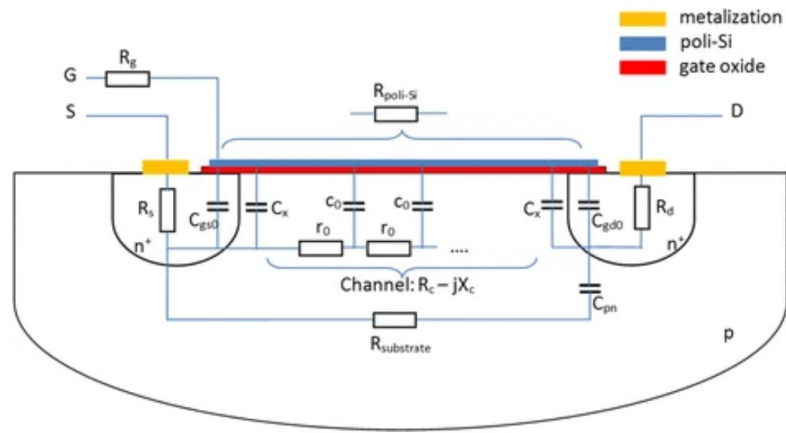
FREQUENCY RESPONSE OF AMPLIFIERS

— magnitude
- - - phase

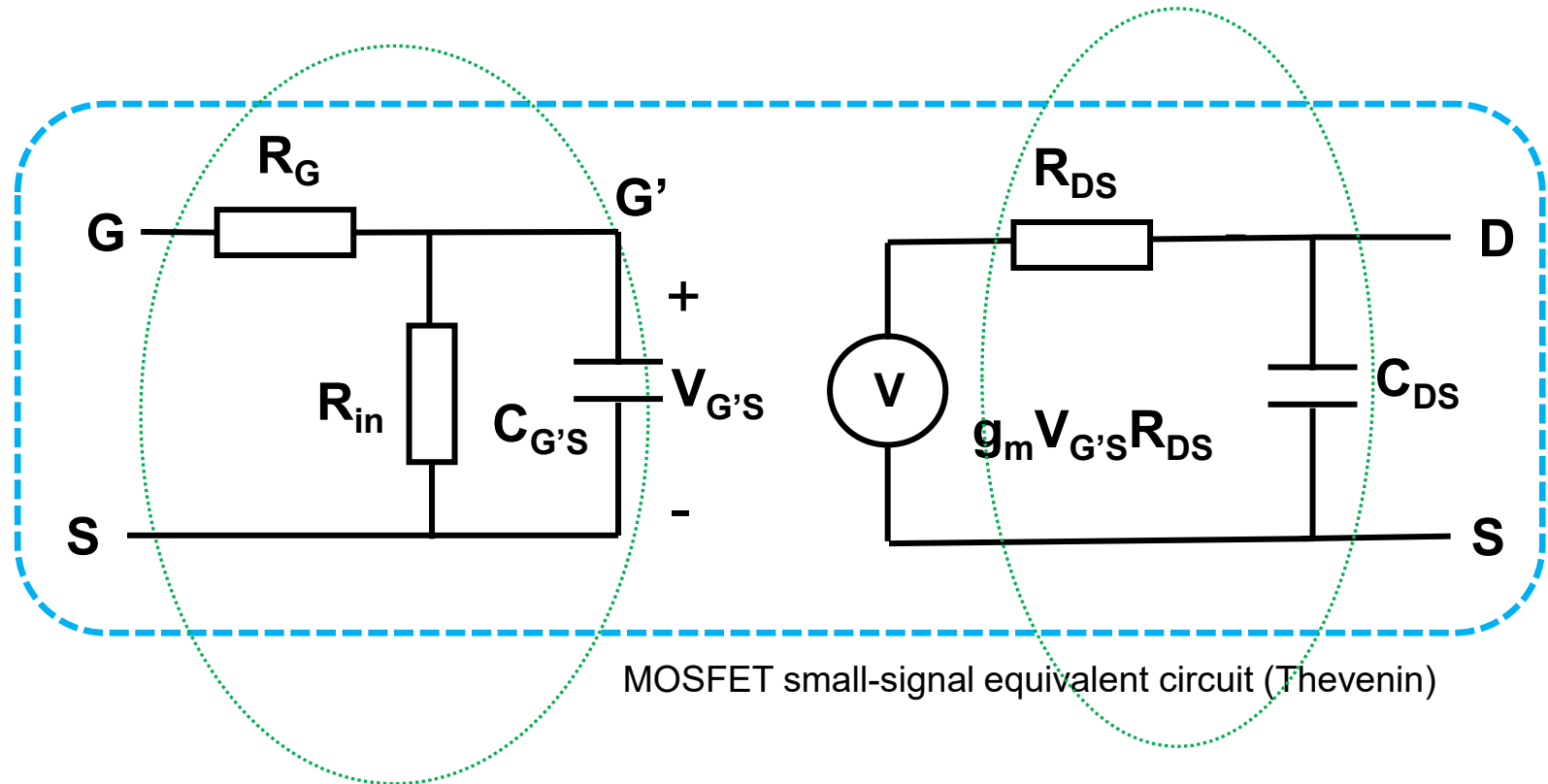


FREQUENCY RESPONSE OF AMPLIFIERS

A MOSFET is not an ideal component and has parasitic capacitances and resistances. The MOSFET with the parasitic can be characterized as follows:



Simplified cross-section of MOSFET



First Low Pass Filter

Second Low Pass Filter

FREQUENCY RESPONSE OF AMPLIFIERS

$$f_1 = 1/(2\pi R_g C_{GS}) \approx 318 \text{ MHz}$$

$$f_2 = 1/(2\pi R_{DS} C_{DS}) \approx 796 \text{ Hz}$$

The corner frequency f_1 is compared to f_2 too high, therefore it can be neglected!

$V_{IN} \approx V_{GS} \rightarrow V_{IN}$ is 1V. In AC analysis, this voltage is indeed infinitely small such that the transistor behaves in a linear fashion. 1V is just a fictitious value used to simplify the calculation of the transfer function.

$$\begin{aligned} \text{Gain}_{DC} &= g_m V_{GS} R_{DS} \\ &= 10\text{mS} \cdot 1\text{M}\Omega = 10000 = 80 \text{ dB} \end{aligned}$$

FREQUENCY RESPONSE OF AMPLIFIERS

The gain-bandwidth GBW product of an electronic component is constant: the gain is reduced, the corner frequency increases vice versa.

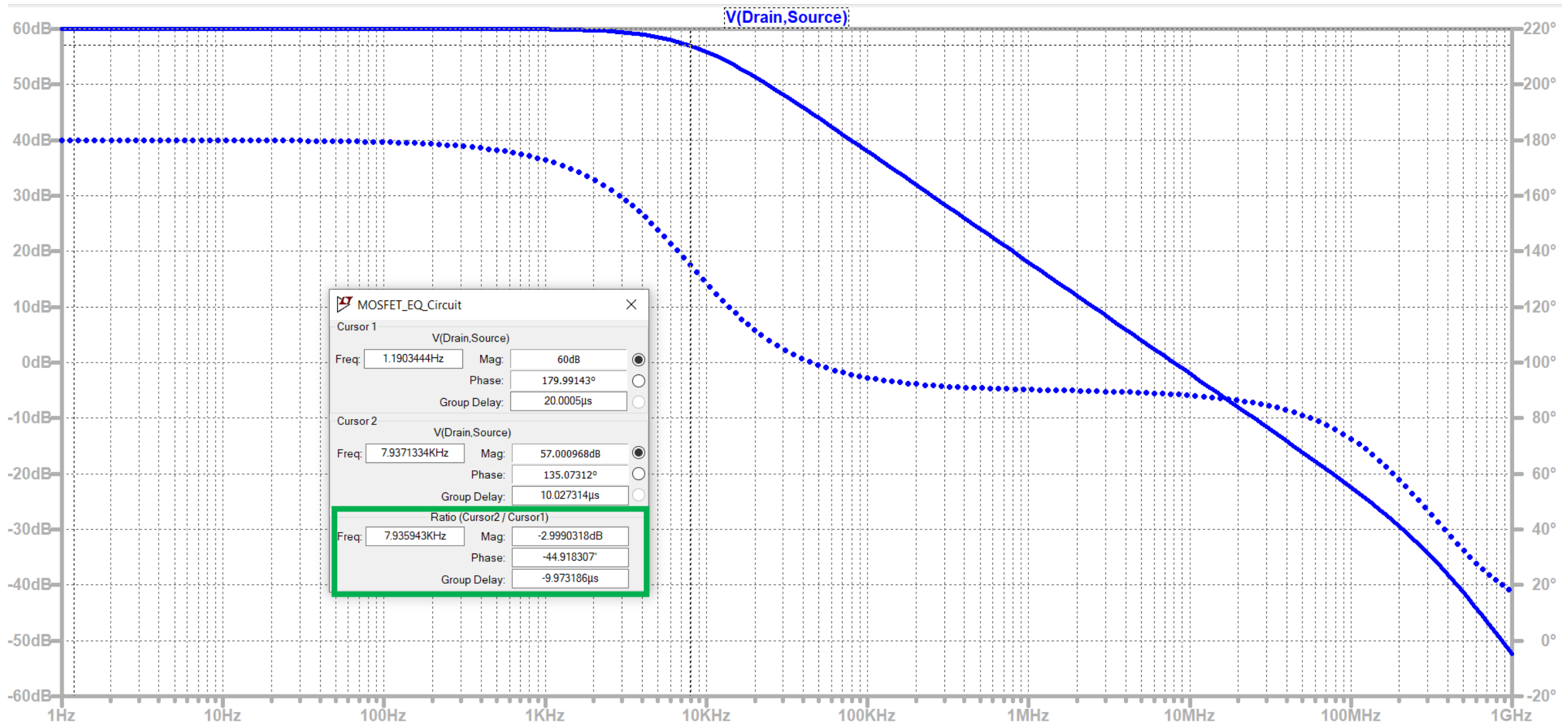
In the previous example:

$$\left. \begin{array}{l} f_2 = 1/(2\pi R_{DS} C_{DS}) \approx 796 \text{ Hz} \\ \text{Gain}_{DC} = 80 \text{ dB} \rightarrow 10000 (10^{80/20}) \end{array} \right\} \text{GBW} = 7.96 \text{ MHz}$$

If we decrease R_{DS} from $1\text{M}\Omega$ to $100\text{k}\Omega$, the corner or the 3-dB frequency is simulated as follows: $\rightarrow$

$$f_2 = 1/(2\pi R_{DS} C_{DS}) \approx 7.94 \text{ kHz}$$

FREQUENCY RESPONSE OF AMPLIFIERS



FREQUENCY RESPONSE OF AMPLIFIERS

According to the simulation and the calculations:

$$\left. \begin{array}{l} f_2 = 7.96 \text{ kHz} \\ \text{Gain}_{\text{DC}} = 60 \text{ dB} \rightarrow 1000 \end{array} \right\} \text{GBW} = 7.96 \text{ MHz}$$

same as in the previous example!

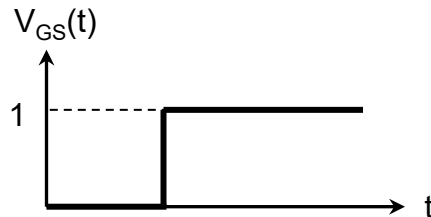
FREQUENCY RESPONSE OF AMPLIFIERS

The small-signal equivalent circuit of a MOSFET is basically a voltage-controlled voltage source with parasitic components (resistance, capacitance, inductance). If we assume that the transistor has a first-order low-pass filter for the sake of simplicity (we have seen previously that for a low-pass filter):

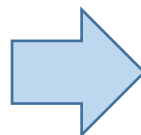
$$\frac{V_{OUT}(s)}{V_{IN}(s)} = \frac{1}{1 + sCR}$$

Inverse Laplace Transform of a low-pass filter:

For a positive step response:



$$\frac{V_{DS}(s)}{V_{GS}(s)} = \frac{V_{DS}(s)}{s} = \frac{1/\tau}{s+1/\tau} \frac{1}{s} = \frac{A}{s} - \frac{B}{s+1/\tau}$$

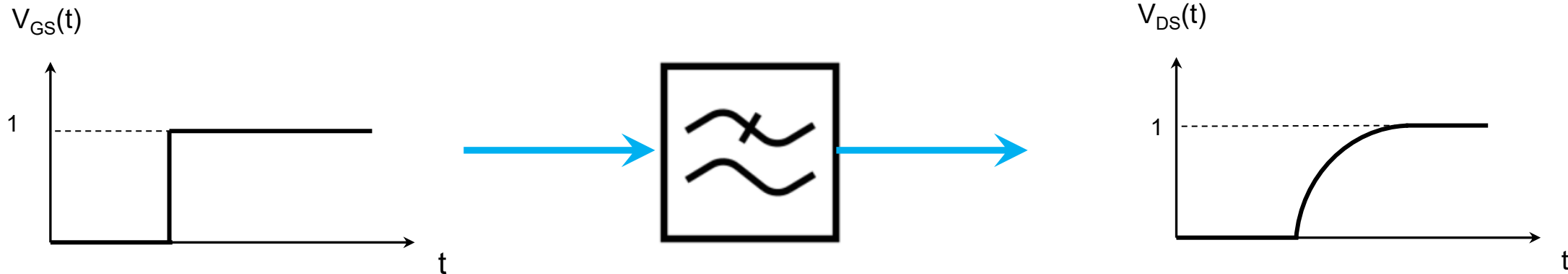


$$A=1, B=-1$$

FREQUENCY RESPONSE OF AMPLIFIERS

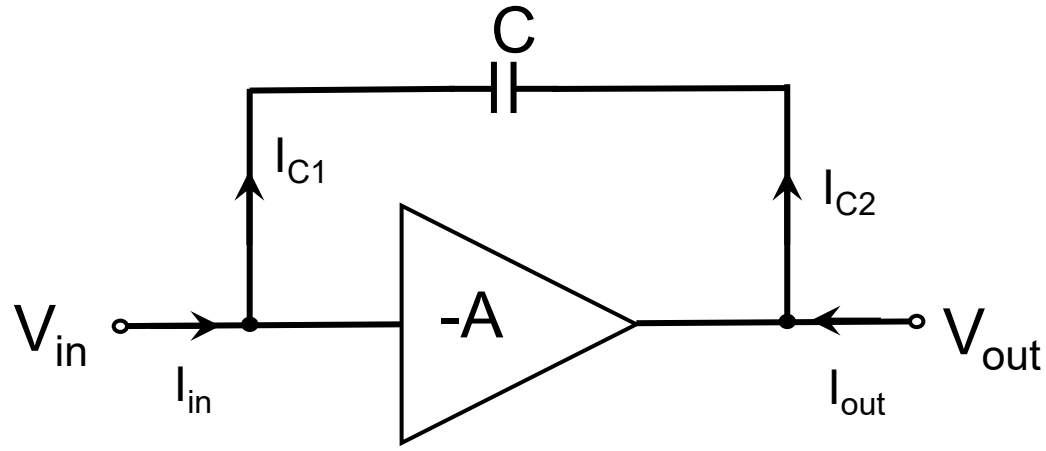
$$V_{DS}(s) = \frac{1}{s} - \frac{1}{1 + 1/\tau}$$

$$\mathcal{L}^{-1}\{V_{DS}(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s} - \frac{1}{1 + 1/\tau}\right\} = 1 - e^{-t/(RC)}$$

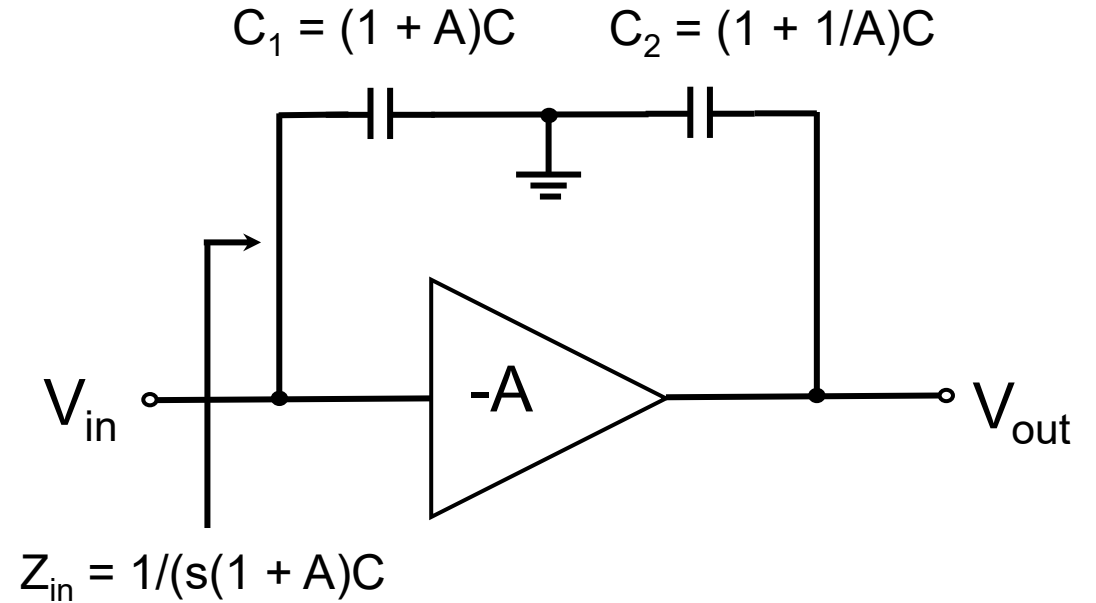


A low-pass filter changes the step response such that at the output the signal has a rise time. The rise time increases with parasitic resistances and capacitances. **The more transistors are involved in a circuit, the more pronounced the effect of the low-pass filter becomes due to an increase in parasitic resistance and capacitance (e.g. amplifiers)!**

FREQUENCY RESPONSE OF AMPLIFIERS — MILLER CAPACITANCE



$\equiv$



Miller Effect causes a reduction of input impedance, reducing at the same time the voltage gain!!!

$$i_{C1} = C \frac{d}{dt} (V_{in} - V_{out})$$

$$i_{C1} = sC (V_{in} - V_{out})$$

$$(V_{out} = -AV_{in})$$

$$i_{C1} = sC (V_{in} - (-AV_{in})) = sC (1 + A) V_{in}$$

$$i_{C1} = sC_1 V_{in}$$

$$sC_1 V_{in} = sC (1 + A) V_{in}$$

$$C_1 = C (1 + A)$$

$$i_{C2} = sC (V_{out} - V_{in})$$

$$i_{C2} = sC \left(V_{out} - \left(-\frac{V_{out}}{A} \right) \right) = sC \left(1 + \frac{1}{A} \right) V_{out}$$

FREQUENCY RESPONSE OF AMPLIFIERS — MILLER CAPACITANCE

$$i_{C1} = C \frac{d}{dt} (V_{in} - V_{out})$$

$$i_{C1} = sC(V_{in} - V_{out})$$

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$$i_{C1} = sC_1 V_{in}$$

$$sC_1 V_{in} = sC(1 + A) V_{in}$$

$$C_1 = C(1 + A)$$

$$i_{C2} = sC(V_{out} - V_{in})$$

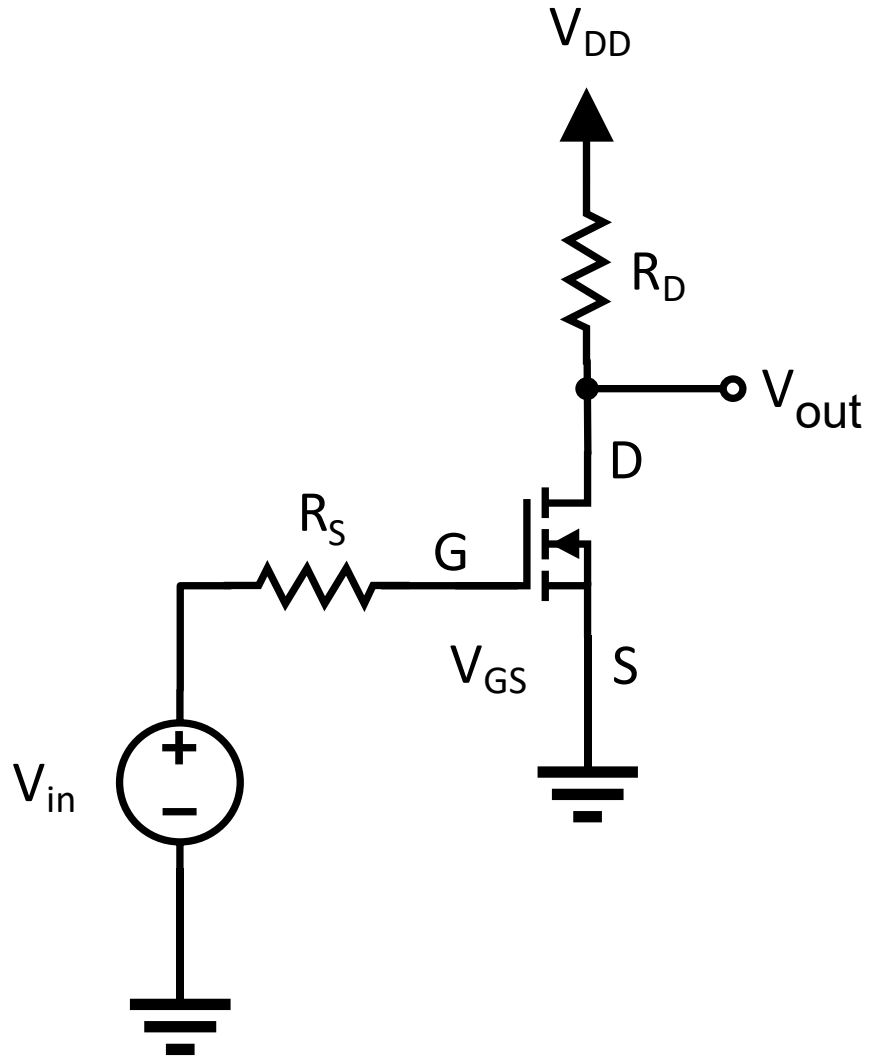
$$i_{C2} = sC \left(V_{out} - \left(-\frac{V_{out}}{A} \right) \right) = sC \left(1 + \frac{1}{A} \right) V_{out}$$

$$i_{out} = sC_2 V_{out}$$

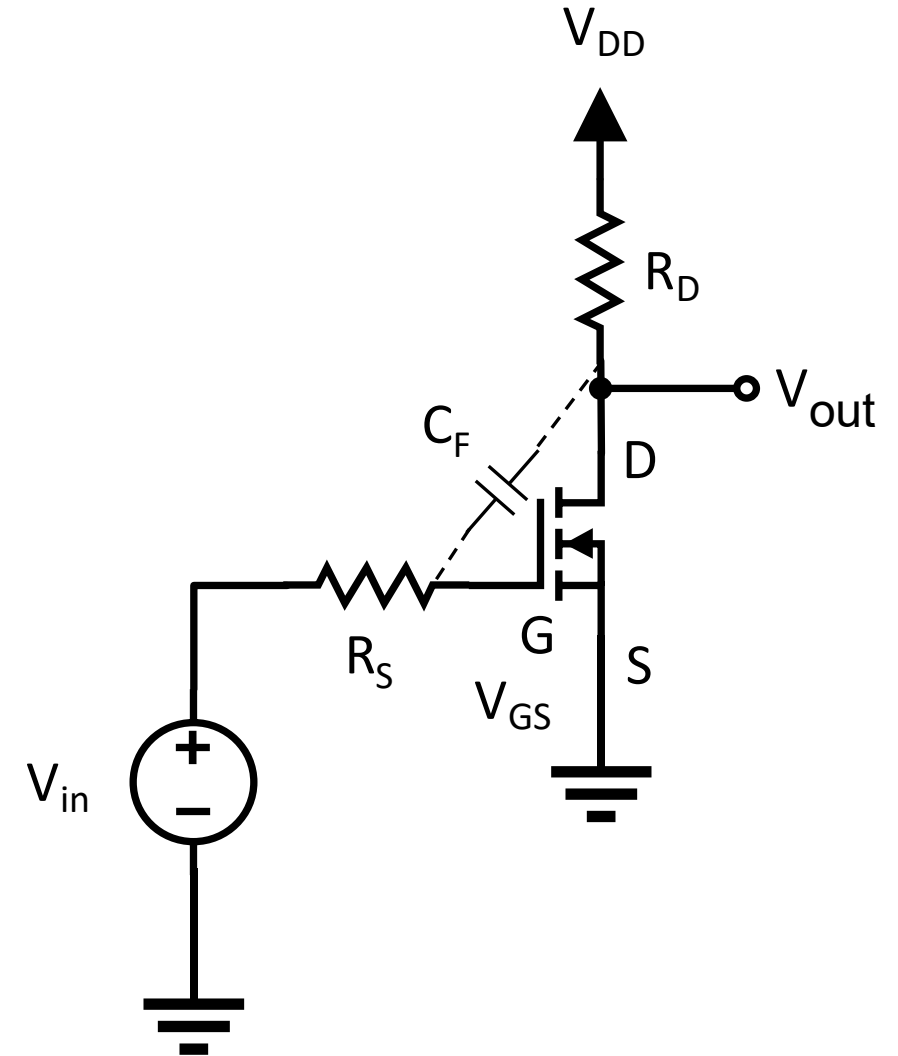
$$sC_2 V_{out} = sC \left(1 + \frac{1}{A} \right) V_{out}$$

$$C_2 = \left(1 + \frac{1}{A} \right) C$$

FREQUENCY RESPONSE OF AMPLIFIERS — MILLER CAPACITANCE

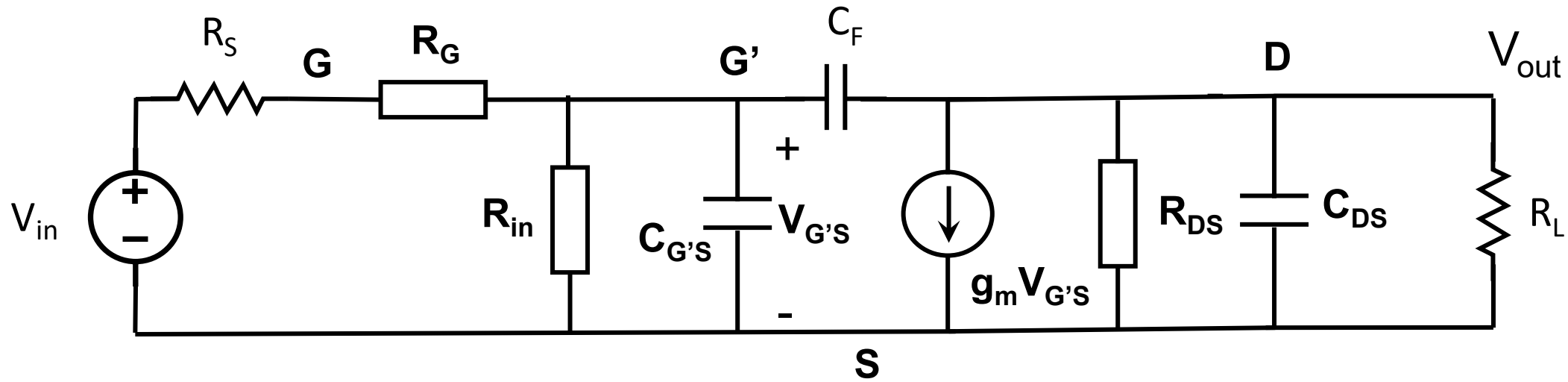


Common-source stage



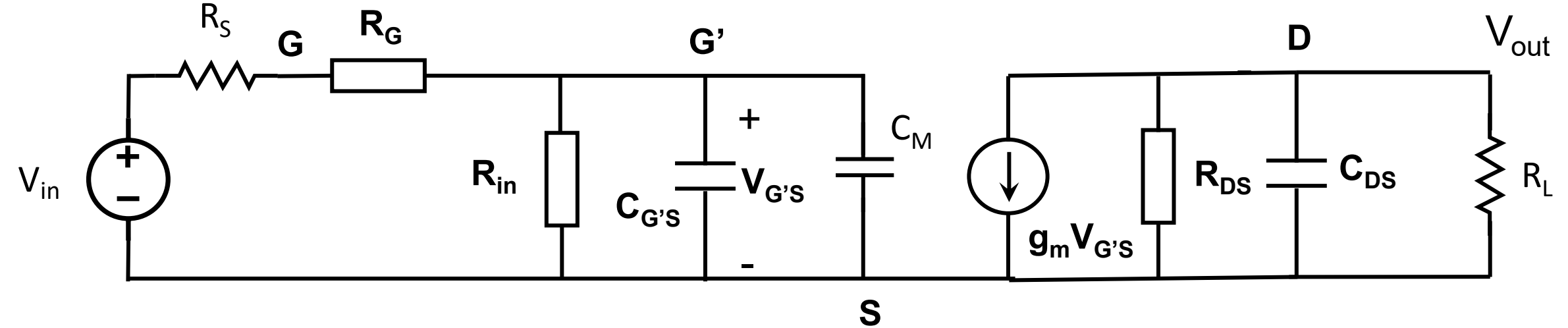
Common-source stage with parasitic Drain-Source Capacitance

FREQUENCY RESPONSE OF AMPLIFIERS — MILLER CAPACITANCE



The physical origin of the Miller capacitance is found in the voltage gain of the circuit. At low frequencies, a small input voltage produces a large output voltage $V_{out} = -g_m R_L V_{in}$ of opposite polarity (negative feedback). This negative feedback together with reduced input impedance leads to a lower gain.

FREQUENCY RESPONSE OF AMPLIFIERS – MILLER CAPACITANCE



The Miller Capacitance C_M is larger than C_f by the factor $(1 + g_m R_2)$.

$$\frac{V_{out}}{V_{in}} = -g_m R_2 \frac{1 - s \frac{C_f}{g_m}}{1 + s \{ R_1 [C_{G's} + C_f (1 + g_m R_2)] + R_2 (C_f + C_{DS}) \} + s^2 R_1 R_2 (C_f C_{G's} + C_{G's} C_{DS} + C_f C_{DS})}$$

Right-Half Plane Zero (RHP): g_m/C_f

where,

$$R_1 = R_S + R_G$$

$$R_2 = R_{DS} \parallel R_L$$

We neglect the impact of R_{in} !

RHP introduces excessive phase shift, thus reduces the phase margin.

FREQUENCY RESPONSE OF AMPLIFIERS – MILLER CAPACITANCE

V_{out} / V_{in} has a positive real zero with magnitude g_m/C_f . This zero stems from the transmission of the signal directly through C_f to the output. The effect of this zero is small except at very high frequencies. This positive real zero can be important in the stability analysis of some operational amplifiers. The denominator shows that the transfer function has two poles, which are usually real and widely separated in practice. If the poles are at p_1 and p_2 , we can write the denominator as

$$D(s) = (1 - s/p_1)(1 - s/p_2) \quad a_1 = -(1/p_1 + 1/p_2) \quad p_1 = -\omega_1 \quad a_2 = R_1 R_2 [C_f(C_{G'S} + C_{DS}) + C_{G'S} C_{DS}]$$

$$D(s) = 1 - s(1/p_1 + 1/p_2) + s^2/(p_1 p_2) \quad a_2 = 1/(p_1 p_2) \quad p_2 = -1/(\omega_1 a_2) \quad p_2 = -1 / [\omega_1 R_1 R_2 (C_f(C_{G'S} + C_{DS}) + C_{G'S} C_{DS})]$$

$$p_1 = - \frac{1}{R_1 [C_{G'S} + C_f(1 + g_m R_2 + R_2/R_1)] + R_2 C_{DS}}$$

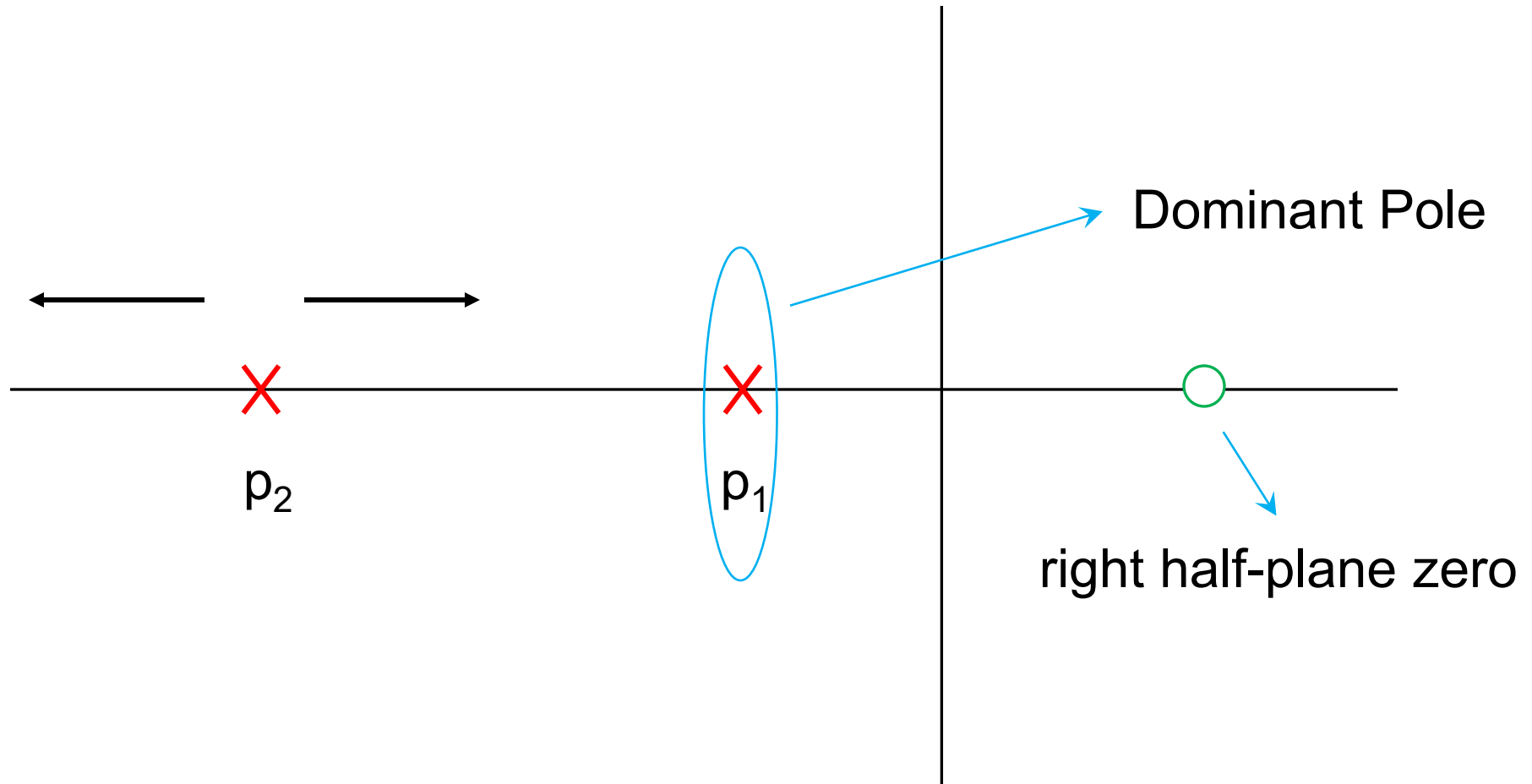
$$p_2 = - \frac{1}{\omega_1 R_1 R_2 [C_f(C_{G'S} + C_{DS}) + C_{G'S} C_{DS}]}$$

for $C_f = 0$ F,

$f_{10} = 63.66$ kHz and $f_{20} = 636.6$ kHz

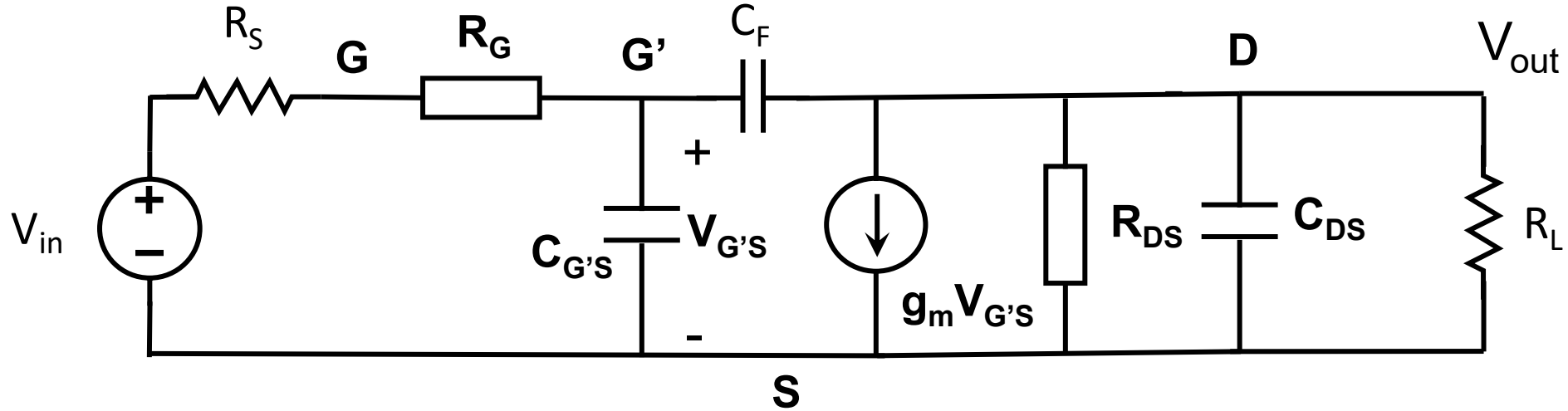
We neglect the impact of R_{in} !

FREQUENCY RESPONSE OF AMPLIFIERS — MILLER CAPACITANCE



The location of the second pole has an impact on the stability of the amplifier (Pole splitting).

EXAMPLE



$$R_1 = R_S + R_G = 100 \text{ k}\Omega$$

$$R_2 = R_{DS} \parallel R_L = 50 \text{ k}\Omega$$

$$C_f = 2 \text{ pF}$$

$$C_{G'S} = 25 \text{ pF}$$

$$C_{DS} = 5 \text{ pF}$$

$$g_m = 4 \text{ mS}$$

$$\omega_1 = 23230 \text{ rad/s, dominant pole} \rightarrow 3.7 \text{ kHz}$$

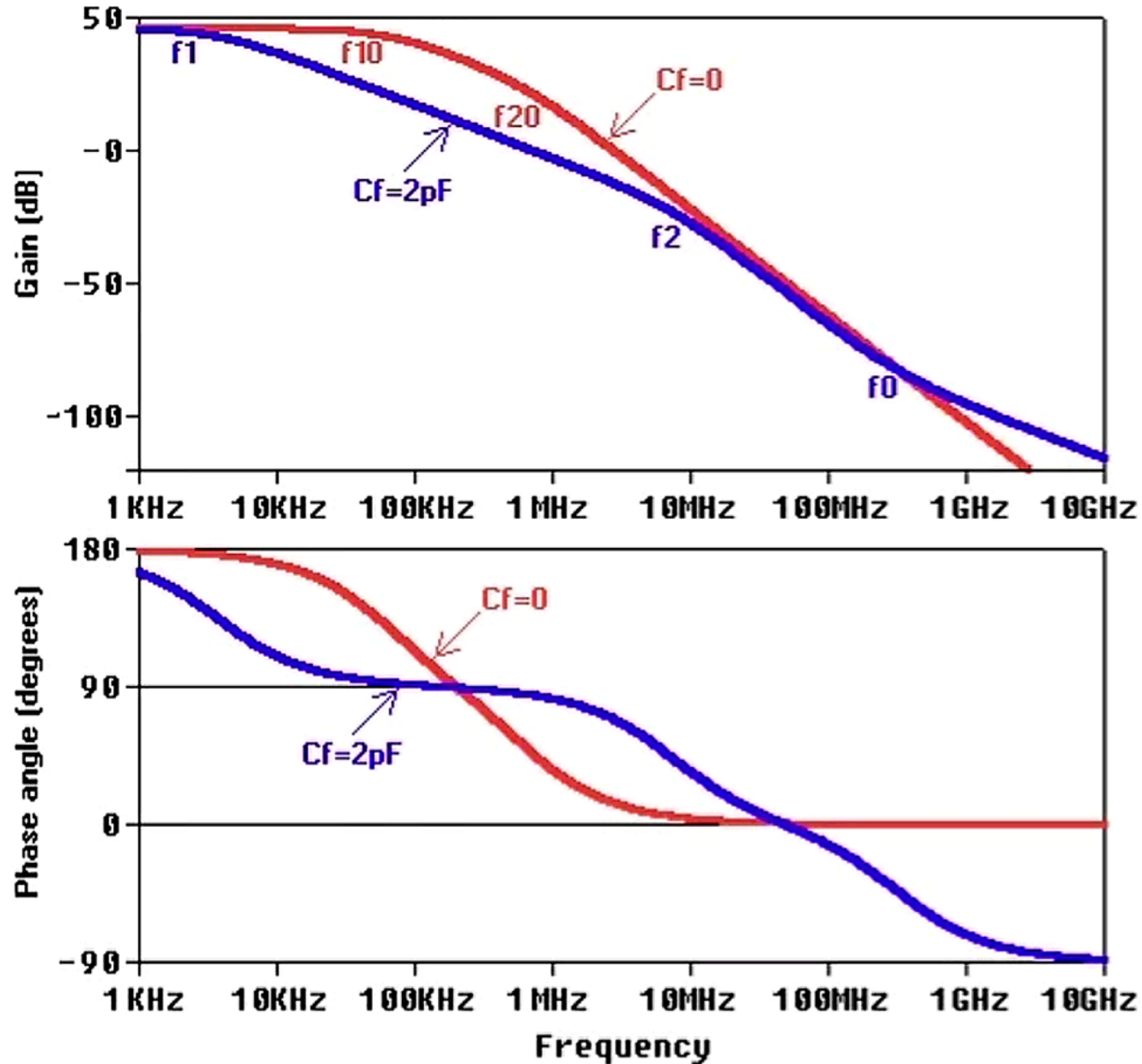
$$\omega_2 = 46.4 \cdot 10^6 \text{ rad/s, dominant pole} \rightarrow 7.38 \text{ MHz}$$

$$\omega_0 = \frac{g_m}{C_f} = \frac{0.004}{2 \cdot 10^{-12}} = 2 \cdot 10^9 \text{ rad/s, zero} \rightarrow 318 \text{ MHz}$$

$$p_1 = -\frac{1}{R_1[C_{G'S} + C_f(1 + g_m R_2 + R_2/R_1)] + R_2 C_{DS}}$$

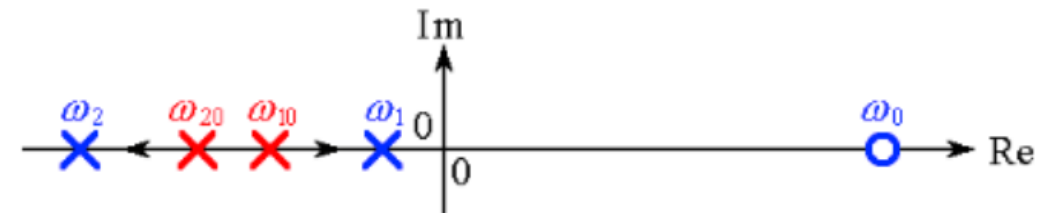
We neglect the impact of R_{in} !

EXAMPLE



In summary, the presence of C_f has a triple effect:

1. It *downshifts* the first pole frequency to 3.7 kHz
2. It *upshifts* the second pole frequency to 7.38 MHz
3. It establishes a *transmission zero* at 318 MHz.



- The stability of amplifiers shall be analyzed in detail.
- The second-pole or the right half plane zero might also have a negative impact on the stability.

EXAMPLE

Consider a common-source MOSFET amplifier where the small-signal parameters are given as follows: the transconductance is $g_m = 2 \text{ mS}$, the output resistance r_o can be taken as infinite, the drain resistor is $R_d = 10 \text{ k}\Omega$, and the signal source has a source resistance $R_s = 1 \text{ k}\Omega$. The MOSFET has a gate–drain capacitance $C_{gd} = 2 \text{ pF}$, while the gate–source capacitance can be neglected. The low-frequency voltage gain of the amplifier is approximately $A_v \approx -g_m \cdot R_d$. Using this information, determine the effect of the gate–drain capacitance through the Miller approximation by finding the equivalent input and output capacitances. Then calculate the dominant pole frequency at the input node. After that, identify the location of the zero in the transfer function and state whether it lies in the right half plane or left half plane.

Low-frequency gain:

$$A_v \approx -g_m \cdot R_d = -0.002 \cdot 10^4 = -20$$

Miller Capacitances:

$$C_{in} = C_{gd} \cdot (1 - A_v) *$$

$$C_{out} = C_{gd} \cdot (1 - 1/A_v)$$

Since $A_v = -20$:

$$C_{in} = 2 \text{ pF} \cdot (1 - (-20)) = 2 \text{ pF} \cdot 21 = 42 \text{ pF}$$

$$C_{out} = 2 \text{ pF} \cdot (1 - (-1/20)) = 2 \text{ pF} \cdot 1.05 = 2.1 \text{ pF}$$

* Here, A_v is already negative since we have a common-source amplifier, therefore,

$C_{in} = C_{gd} \cdot (1 - A_v)$ and NOT $C_{in} = C_{gd} \cdot (1 + A_v)$!!!

Dominant input pole:

Input node sees R_s and C_{in} :

$$\omega_{p,in} = 1 / (R_s \cdot C_{in})$$

$$\omega_{p,in} = 1 / (10^3 \cdot 42 \times 10^{-12})$$

$$\omega_{p,in} \approx 2.38 \cdot 10^7 \text{ rad/s}$$

$$f_{p,in} = \omega / (2\pi) \approx 3.8 \text{ MHz}$$

Zero Location:

For a common-source stage with C_{gd} , the zero is:

$$\omega_z = g_m / C_{gd}, \omega_z = 0.002 / (2 \times 10^{-12}) = 10^9 \text{ rad/s}$$

This is a right-half-plane (RHP) zero.

References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.